

Thesis Revisions

Acoustic Information Retrieval for Interactive Sound Rendering in Virtual Environments

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3rd February, 2025

No.	Pages ¹	Revision	Action Taken
1.	iii	Abstract revision: emphasise the research's originality and importance by clearly outlining the pipeline's purpose and concisely summarising the key findings for a stronger impact	Clarified ambiguity between AR and MR terms. Novel contributions explicitly stated in the abstract. Rewritten and refactored paragraphs to improve clarity and overall conciseness of the abstract, fitting it into a single page.
2.	t.m.	Language Consistency: Ensure the dissertation uses British English consistently throughout.	Checked and corrected instances of grammar errors, spelling mistakes, and unclear sentences throughout the manuscript.
3.	t.m.	Terminology Standardisation: Maintain uniform terminology across the document, particularly when discussing AR, MR, and XR technologies.	Clarified the wrong use of the language and terminology around XR, MR, and AR across the relevant chapters and in the abstract.
4.	74	Dataset Description: Provide a clear explanation of the dataset, including any limitations and the segmentation methods applied.	Added methodology details on the dataset generation for the camera-based system to increase reproducibility and stated limitations of the trained models around their generalisability at the top of the Chapter.
5.	t.m.	Figure Citation and Organisation: Ensure that all figures are correctly cited and logically integrated within the text. Re-arrange figures where needed to improve the document's flow.	All Figures are correctly referenced. Figure 2.1 was resized to improve the document's flow.
6.	7, 33	Avoid Figure Duplication: Ensure each figure is unique within the thesis and avoid repeating the same visuals across sections.	Figure 1.1 was removed from Chapter 6, kept in Chapter 1, and correctly referenced. Figure 2.12 was removed from Chapter 5, kept in Chapter 2, and correctly referenced.
7.	81, 79	Table 4.3 Clarity: Organise the results in Table 4.3 to enhance readability and prevent potential confusion.	Table 4.3 in question has been re-rendered, dividing it into two parts and making it clear that it shows two scenes. The caption has been updated to further clarify how results are organised visually. Table 4.2 has been updated to include a more descriptive caption and labelled axes to show frequency values.
8.	87	Figure 4.10 Explanation: Provide additional clarification on Figure 4.10, specifically regarding the predicted class labels for improved understanding.	The three Figures in question have been grouped into one, using subfigures, adding a caption to clarify the process visualised by them. The Figure is now standalone (Figure 4.9 in the redacted document) and there are in-text references that explain the engineering process.
9.	t.m.	There are some writing issues in the thesis that need to be fixed. One of the examples are the sentences that are not well structured and do not make much sense (e.g. missing a verb) – see the comments in the PDF.	Improvements on the writing, paragraph structure, and sentences are made following all annotations provided on the PDF. This was carried out on all Chapters.

¹t.m. means throughout the manuscript. References to page numbers, Figures, and Tables in this document refer to the redacted manuscript.

10.	1–3	Section 1.1. Define and introduce presence, immersion, and embodiment, and then link them to audio, and then add the interaction and interfaces.	As per discussion, the term embodiment is not relevant to the work and has been removed from the manuscript. Sections 1.1.1 and 1.1.2 have been collapsed into one, making their writing more concise and improving their readability.
11.	5	Section 1.1.5 should be rewritten. There are some writing issues, but also it talks mostly about XR and briefly mentions the audio interaction system – this should be balanced better.	Following the guidance from the PDF and similarly to Revision 10, Sections 1.1.4 and 1.1.5 have been collapsed into one, making their writing more concise and improving their readability.
12.	10	Make the contributions (Section 1.7) a bit clearer, using more consistent terminology and potentially listing them chronologically.	As per discussions with the committee, Key Performance indicators were removed from Section 1.4. Section 1.7 has been improved, making the contributions concise and easier to read.
13.	t.m.	Reconsider the sizes of the figures in the thesis. Some of them are unnecessarily too large.	This has been addressed as part of Revision 5.
14.	t.m.	I am slightly concerned about the repetition and redundancy in the thesis. Some concepts/definitions/explanations are repeated multiple times in the thesis which could have been avoided. It is good to link sections and chapters, but you do not need to explain same things multiple times. See the PDF for specific examples.	This has been addressed as part of Revisions 6 and 9. All specific examples annotated in the PDF are addressed.
15.	80–83	Some details on the method and the setup are provided in the results (e.g. Section 4.2.4), although these should be mentioned earlier.	Section 4.2.4 and 4.3.3 have been edited, moving all methodology aspects to earlier Sections or any discussion about results to appropriate Discussion Sections. The Chapter has been improved to clarify unclear parts (rewritten Section 4.3.2, updated graphics in Figure 4.11), as per recommendations provided.
16.	107	I am slightly bothered with calling Section 5.2.8 Subjective Evaluation. A perceptual model CDPAM, trained on human judgements is used, but it is still algorithmic. This distinction is important because it shifts the evaluation from a truly subjective approach (direct human feedback) to an indirect or model-based approach that emulates human perceptual evaluation. A more precise term might be "perceptual model-based evaluation" or "model-based perceptual similarity assessment". This terminology would clarify that, while the model is trained on subjective data, the evaluation is not directly subjective but rather a proxy of subjective assessment.	Section 5.1.8 in Chapter 5 has been updated to clarify that the data is extracted using Learned Similarity Metrics and reasoning around the used of a learned metric is added to provide the reader with context. Honouring the PDF annotations, the rest of the Chapter has been improved.